Emre Gurpinar, Ph.D.

Senior Staff - Associate Technical Fellow · Sikorsky Aircraft

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Education

The University of Nottingham — *Nottingham, UK*
Ph.D. in Electrical and Electronics Engineering — *Jun. 2013 - Feb. 2017*
- Advisor: Prof. Alberto Castellazzi
- Thesis Title: Wide-Bandgap Semiconductor Based Renewable Power Converters

The University of Manchester — *Manchester, UK*
M.Sc. in Electrical Energy Conversion Systems — *Oct. 2009 - Oct. 2010*
- Advisor: Prof. Andrew J. Forsyth
- Thesis Title: Design, Construction, and Assessment of a High-Frequency DC-DC Converter Using Silicon Carbide Power Devices

Istanbul Technical University — *Istanbul, Turkey*
B.Sc. in Electrical Engineering — *Sep. 2005 - Jun. 2009*

Professional Experience

Sikorsky Innovations, Sikorsky Aircraft — *Connecticut, USA*
Senior Staff - Associate Technical Fellow — *Mar. 2022 - Present*
- Leading power electronics component and system development for electrified vertical take-off and landing (eVTOL) experimental demonstrators and future electric aircraft propulsion solutions.
- Recognized subject matter expert as an Associate Technical Fellow in power semiconductor devices, power electronics and electric propulsion systems.

Electric Drives Research Group, Oak Ridge National Laboratory — *Tennessee, USA*
Research Staff — *May. 2018 - Mar. 2022*
- Led multiple research projects as a principal investigator on wide-bandgap device characterization, semiconductor packaging, and multi-physics analysis and optimization.
- Supported high power static and dynamic wireless charging demonstrators for electric vehicle charging.

Electric Drives Research Group, Oak Ridge National Laboratory — *Tennessee, USA*
Postdoctoral Research Associate — *May. 2017 - May. 2018*
- Conducted analysis and experimental characterization of wide-bandgap devices for electric vehicles.

The University of Nottingham — *Nottingham, UK*
Graduate Student — *Jun. 2013 - Mar. 2017*
- Worked on characterization, modeling, and implementation of high voltage wide-bandgap power semiconductors in renewable energy applications at Power Electronics, Control, and Machines Research Group.

Advanced Technology Group, GE Power Conversion — *Rugby, UK*
Research and Development Engineer — *Dec. 2011 - Jun. 2013*
- Analyzed, modeled, and tested high-power electric drive systems for wind turbine and marine applications.

Department of Systems Engineering, The University of Reading *Reading, UK*
RESEARCH AND DEVELOPMENT ENGINEER *Oct. 2010 - Dec. 2011*

- Developed analytical and finite element electromagnetic models of slot-less permanent magnet motors for residential ventilation systems.

Grants, Awards, & Scholarships

Grants

2021	**High Power Density Inverter for MD & HD Electric Trucks**, Cummins & US DOE Vehicle Technologies Office	*$650,000*
2019-2022	**Highly Integrated Power Module**, US DOE Vehicle Technologies Office	*$2,232,000*
2019	**Integrated Drive System for Electric Vehicles**, Toyota North America R&D	*$250,000*
2018	**DC Link Capacitor Characterization for EV Systems**, Rogers Corporation	*$120,000*
	Advanced Multiphysics Integrations and Designs, US DOE Vehicle Technologies Office	*$700,000*
2016	**Multi-level Inverter Analysis**, The University of Nottingham	*£20,000*

Awards

2019	**Outstanding Paper Award**, ASME International Technical Conference on Packaging and Integration of Electronic and Photonic Microsystems (InterPACK)

Scholarships

2014-2017	**Faculty of Engineering Scholarship**, The University of Nottingham	*£48,000*

Mentoring

At Oak Ridge National Laboratory, I supervised two post-doctoral researchers on multi-physics modeling and optimization of semiconductor packaging as part of my research grants. In addition, I mentored two undergraduate summer interns from the University of Tennessee, Knoxville. Finally, I co-supervised a Ph.D. student from the Georgia Institute of Technology on heat sink optimization for high-power semiconductor modules.

At the University of Nottingham, I mentored one undergraduate and one graduate student for their thesis related to resonant gate driver design and high-frequency DC/DC converters. I also supervised a visiting undergraduate summer intern in embedded programming for high-frequency power converter control.

Invited Talks

B. Ozpineci, **E. Gurpinar**, and S. Chowdhury, Tutorial on ”Wide-bandgap Devices for Automotive Applications”, IEEE Workshop on Wide Bandgap Power Devices and Applications (WiPDA), 7 November 2021.

E. Gurpinar, B. Ozpineci, ”High-Performance Heat Sink Design for WBG Power Modules using Genetic Algorithms”, IEEE Power Electronics Society DMC Webinar Series, 25 March 2021.
https://resourcecenter.ieee-pels.org/education/webinars/pelsweb032521v

B. Ozpineci, **E. Gurpinar**, M. Chinthavali, Z. Wang, ”Application of Wide-Bandgap Semiconductors for Electric Vehicles”, IEEE Industry Applications Society Webinar Series, 15 March 2018.
https://resourcecenter.ias.ieee.org/education/technical-webinars/IASWEB0032.html

E. Gurpinar ”Wide-Bandgap Devices in Renewable Power Converters” PhD Candidate Meeting - IEEE International Power Electronics Conference, Hiroshima, Japan, 18 May 2014.

Outreach & Professional Development

Professional Service

2023-2024 **International Symposium on Power Semiconductor Devices and ICs**, Technical Program Committee Member

2020-2024 **ASME InterPACK Conference**, Power Electronics Track Co-Chair

2020-2023 **IEEE Energy Conversion Congress and Exposition**, Power Semiconductor Devices, Passive Components, Packaging, and Materials Subcommittee Member

2020-2023 **IEEE Energy Conversion Congress and Exposition**, Control, Modeling, and Optimization of Power Converters Subcommittee Member

2020 **ASME InterPACK Conference**, Technical Session Chair

2020 **IEEE Workshop on Wide Bandgap Power Devices and Applications in Asia**, Technical Program Committee Member

2020 **IEEE Cybersecurity for Power Electronics (CyberPELS)**, Publication Chair

2018 **IEEE Transportation Electrification Conference and Expo**, Technical Session Chair

Peer Review and Editorial Positions

Associate Editor - IEEE Journal of Emerging and Selected Topics in Power Electronics

Reviewer - IEEE Transactions on Power Electronics

Reviewer - IEEE Transactions on Industry Applications

Reviewer - IEEE Journal of Emerging and Selected Topics in Power Electronics

Professional Memberships

Senior Member - IEEE

Member - IEEE Power Electronics Society

Member - IEEE Industry Applications Society

Teaching Experience

2014-2016 **Resident Tutor**, Sherwood Hall, The University of Nottingham

2014-16 **Electronic Circuit Design**, Teaching Assistant, The University of Nottingham

2015-16 **Electro-mechanical Systems**, Teaching Assistant, The University of Nottingham

Publications in Reverse Chronological Order

Journal Articles

[18] L. Xue, V. Galigekere, **E. Gurpinar**, G. Su, S. Chowhdhury, M. Mohammad, O. Onar, ”Modular Power Electronics Approach for High Power Dynamic Wireless Charging System,” in *IEEE Transactions on Transportation Electrification* vol. 10, no. 1, pp. 976-988, March 2024.

[17] R. Sahu, **E. Gurpinar**, and B. Ozpineci. ”Liquid-Cooled Heat Sink Optimization for Thermal Imbalance Mitigation in Wide-Bandgap Power Modules.” *ASME. J. Electron. Packag.* June 2022

[16] S. Chowdhury, **E. Gurpinar** and B. Ozpineci, ”Capacitor Technologies: Characterization, Selection, and Packaging for Next-Generation Power Electronics Applications,” in *IEEE Transactions on Transportation Electrification*, vol. 8, no. 2, pp. 2710-2720, June 2022.

[15] **E. Gurpinar**, R. Sahu and B. Ozpineci, ”Heat Sink Design for WBG Power Modules Based on Fourier Series and Evolutionary Multi-Objective Multi-Physics Optimization,” in *IEEE Open Journal of Power Electronics*, vol. 2, pp. 559-569, 2021.

[14] I. Husain, B. Ozpineci, M.S. Islam, **E. Gurpinar**, G. Su, W. Yu, S. Chowdhury, L. Xue, D. Rahman and R. Sahu, ”Electric Drive Technology Trends, Challenges, and Opportunities for Future Electric Vehicles”, in *Proceedings of the IEEE*, vol. 109, no. 6, pp. 1039-1059, June 2021.

[13] **E. Gurpinar**, S. Chowdhury, B. Ozpineci and W. Fan, ”Graphite Embedded High Performance Insulated Metal Substrate for Wide-bandgap Power Modules”, *IEEE Transactions on Power Electronics*, vol. 36, no. 1, pp. 114-128, Jan. 2021.

[12] J. Garcia, S. Saeed, **E. Gurpinar** and A. Castellazzi, ”A Study of Integrated Signal and Power Transfer for Compact Isolated SiC MOSFET Gate-Drivers” in Electronics, 2021, 10, 159.

[11] **E. Gurpinar**, B. Ozpineci and S. Chowdhury ”Design, Analysis, Comparison, and Experimental Validation of Insulated Metal Substrates for High-Power Wide-Bandgap Power Modules”, *ASME Journal of Electronic Packaging*, December 2020; 142(4): 041107.

[10] **E. Gurpinar**, B. Ozpineci, ”Transient Pulse Based Impedance Characterization of DC Link Capacitor for EV Systems”, *IET Power Electronics*, vol. 13, no. 1, pp. 127-132, 7 1 2020.

[9] A. Castellazzi, **E. Gurpinar**, Z. Wang, A. H. Suliman, and P. Garcia Fernandez ”Impact of Wide-Bandgap Technology on Renewable Energy and Smart-Grid Power Conversion Applications Including Storage”, *Energies*, 2019, 12, 4462

[8] J. Pries, **E. Gurpinar**, L. Tang and T. A. Burress, ”Continuum Modeling of Inductor Magnetic Hysteresis and Eddy Currents in Resonant Circuits,” in *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 7, no. 3, pp. 1703-1714, Sept. 2019.

[7] J. Garcia, S. Saeed, **E. Gurpinar**, A. Castellazzi and P. Garcia, ”Self-Powering High Frequency Modulated SiC Power MOSFET Isolated Gate Driver,” in *IEEE Transactions on Industry Applications*, vol. 55, no. 4, pp. 3967-3977, July-Aug. 2019.

[6] **E. Gurpinar** and A. Castellazzi, ”Tradeoff Study of Heat Sink and Output Filter Volume in a GaN HEMT Based Single-Phase Inverter,” in *IEEE Transactions on Power Electronics*, vol. 33, no. 6, pp. 5226-5239, June 2018.

[5] **E. Gurpinar**, F. Iannuzzo, Y. Yang, A. Castellazzi and F. Blaabjerg, ”Design of Low-Inductance Switching Power Cell for GaN HEMT Based Inverter,” in *IEEE Transactions on Industry Applications*, vol. 54, no. 2, pp. 1592-1601, March-April 2018.

[4] J. Li, A. Castellazzi, M. A. Eleffendi, **E. Gurpinar**, C. M. Johnson and L. Mills, ”A Physical RC Network Model for Electrothermal Analysis of a Multichip SiC Power Module,” in *IEEE Transactions on Power Electronics*, vol. 33, no. 3, pp. 2494-2508, March 2018.

[3] **E. Gurpinar** and A. Castellazzi, ”Single-Phase T-Type Inverter Performance Benchmark Using Si IGBTs, SiC MOSFETs, and GaN HEMTs,” in *IEEE Transactions on Power Electronics*, vol. 31, no. 10, pp. 7148-7160, Oct. 2016.

[2] **E. Gurpinar**, Y. Yang, F. Iannuzzo, A. Castellazzi and F. Blaabjerg, ”Reliability-Driven Assessment of GaN HEMTs and Si IGBTs in 3L-ANPC PV Inverters,” in *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 4, no. 3, pp. 956-969, Sept. 2016.

[1] D. Barater, C. Concari, G. Buticchi, **E. Gurpinar**, D. De and A. Castellazzi, ”Performance Evaluation of a Three-Level ANPC Photovoltaic Grid-Connected Inverter With 650-V SiC Devices and Optimized PWM,” in *IEEE Transactions on Industry Applications*, vol. 52, no. 3, pp. 2475-2485, May-June 2016.

Conference Papers

[27] V. P. Galigekere, **E. Gurpinar**, S. Chowdhury, J. Pries, ”Suppression of Parasitic Voltage Oscillation in Power Modules by Turn-Off Current Zero Placement”, *2023 IEEE Energy Conversion Congress and Exposition (ECCE)*, Nashville, TN, USA, 2023, pp. 5485-5490

[26] N. Rajagopal, **E. Gurpinar**, B. Ozpineci and C. DiMarino, ”Electro-Thermal Optimization of Common-Mode Screen for Organic Substrate-Based SiC Power Module,” *2022 IEEE Energy Conversion Congress and Exposition (ECCE)*, Detroit, MI, USA, 2022, pp. 1-8.

[25] H. Barua, **E. Gurpinar**, L. Xue, and B. Ozpineci ”Comparative Analysis of Direct and Indirect Cooling of Wide-Bandgap Power Modules and Performance Enhancement of Jet Impingement-Based Direct Substrate Cooling.” *ASME 2020 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems.* Garden Grove, California, USA. October 25–27, 2022.

[24] L. Xue, V. Galigekere, G. Su, R. Zheng, M. Mohammad, **E. Gurpinar**, S. Chowdhury, and O. Onar, ”Design and Analysis of a 200 kW Dynamic Wireless Charging System for Electric Vehicles,” *2022 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Houston, TX, USA, 2022, pp. 1096-1103.

[23] L. Xue, V. Galigekere, **E. Gurpinar**, G. Su and O. Onar, ”Modular Design of Receiver Side Power Electronics for 200 kW High Power Dynamic Wireless Charging System,” *2021 IEEE Transportation Electrification Conference & Expo (ITEC)*, 2021, pp. 744-748.

[22] S. Chowdhury, **E. Gurpinar** and B. Ozpineci, ”Electrical Characterization of Insulated Metal Substrate-Based SiC Power Module for Traction Application”, *2020 IEEE Energy Conversion Congress and Exposition (ECCE)*, Detroit, MI, 2020, pp. 195-202.

[21] R. Sahu, **E. Gurpinar** and B. Ozpineci, ”Fourier Analysis-based Evolutionary Optimization of Liquid-cooled Heat Sinks”, *2020 IEEE Energy Conversion Congress and Exposition (ECCE)*, Detroit, MI, 2020, pp. 4017-4023.

[20] R. Sahu, **E. Gurpinar** and B. Ozpineci, ”Liquid-Cooled Heat Sink Optimization for Thermal Imbalance Mitigation in Wide-Bandgap Power Modules.” *ASME 2020 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems.* Virtual, Online. October 27–29, 2020.

[19] **E. Gurpinar**, J.P. Spires, B, Ozpineci and W. Fan, ”Analysis and Evaluation of Thermally Annealed Pyrolytic Graphite Heat Spreader for Power Modules,” *2020 IEEE Applied Power Electronics Conference and Exposition (APEC)*, New Orleans, LO, 2020, pp. 2741-2747.

[18] S. Chowdhury, **E. Gurpinar**, G. Su, T. Raminosoa, T. A. Burress and B. Ozpineci, ”Enabling Technologies for Compact Integrated Electric Drives for Automotive Traction Applications,” *2019 IEEE Transportation Electrification Conference and Expo (ITEC)*, Detroit, MI, USA, 2019, pp. 1-8.

[17] **E. Gurpinar**, S. Chowdhury, and B. Ozpineci, ”Design, Analysis and Comparison of Insulated Metal Substrates for High Power Wide-Bandgap Power Modules.”, *ASME 2019 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems.* Anaheim, California, USA. October 7–9, 2019.

[16] M. Valente, F. Iannuzzo, Y. Yang and **E. Gurpinar**, ”Performance Analysis of a Single-phase GaN-based 3L-ANPC Inverter for Photovoltaic Applications,” *2018 IEEE 4th Southern Power Electronics Conference (SPEC)*, Singapore, Singapore, 2018, pp. 1-8.

[15] **E. Gurpinar** et al., ”SiC MOSFET-Based Power Module Design and Analysis for EV Traction Systems,” *2018 IEEE Energy Conversion Congress and Exposition (ECCE)*, Portland, OR, 2018, pp. 1722-1727.

[14] A. Castellazzi, A. Fayyaz, **E. Gurpinar**, A. Hussein, J. Li and B. Mouawad, ”Multi-Chip SiC MOSFET Power Modules for Standard Manufacturing, Mounting and Cooling,” *2018 International Power Electronics Conference (IPEC-Niigata 2018 -ECCE Asia)*, Niigata, 2018, pp. 130-136.

[13] **E. Gurpinar** and B. Ozpineci, ”Loss Analysis and Mapping of a SiC MOSFET Based Segmented Two-Level Three-Phase Inverter for EV Traction Systems,” *2018 IEEE Transportation Electrification Conference and Expo (ITEC)*, Long Beach, CA, 2018, pp. 1046-1053.

[12] J. Garcia, **E. Gurpinar**, A. Castellazzi and P. Garcia, ”Self-supplied isolated gate driver for SiC power MOSFETs based on bi-level modulation scheme,” *2017 IEEE Energy Conversion Congress and Exposition (ECCE)*, Cincinnati, OH, 2017, pp. 5101-5106.

[11] J. Garcia, **E. Gurpinar** and A. Castellazzi, ”High-frequency modulated secondary-side self-powered isolated gate driver for full range PWM operation of SiC power MOSFETs,” *2017 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Tampa, FL, 2017, pp. 919-924.

[10] **E. Gurpinar** and A. Castellazzi, ”600 V normally-off p-gate GaN HEMT based 3-level inverter,” *2017 IEEE 3rd International Future Energy Electronics Conference and ECCE Asia (IFEEC 2017 - ECCE Asia)*, Kaohsiung, 2017, pp. 621-626.

[9] S. Chowdhury, **E. Gurpinar** and A. Castellazzi, ”Full SiC version of the EDA5 inverter,” *2017 IEEE 3rd International Future Energy Electronics Conference and ECCE Asia (IFEEC 2017 - ECCE Asia)*, Kaohsiung, 2017, pp. 406-411.

[8] **E. Gurpinar**, F. Iannuzzo, Y. Yang, A. Castellazzi, F. Blaabjerg, ”Ultra-low inductance design for a GaN HEMT Based 3L-ANPC Inverter”. in *IEEE Energy Conversion Congress & Expo (ECCE 2016)*, 18-22 Sept 2016, Milwaukee, Wisconsin.

[7] **E. Gurpinar** and A. Castellazzi, ”Novel Multilevel Hybrid Inverter Topology with Power Scalability” in *42^{nd} Annual Conference of IEEE Industrial Electronics Society (IECON 2016)*, 24-27 Oct 2016, Florence.

[6] **E. Gurpinar** and A. Castellazzi, "SiC and GaN based BSNPC inverter for photovoltaic systems," *Power Electronics and Applications (EPE'15 ECCE-Europe)*, 2015 17th European Conference on, Geneva, 2015, pp. 1-10.

[5] D. Barater, C. Concari, G. Buticchi, **E. Gurpinar**, D. De and A. Castellazzi, "Performance evaluation of a 3-level ANPC photovoltaic grid-connected inverter with 650V SiC devices and optimized PWM," *2014 IEEE Energy Conversion Congress and Exposition (ECCE)*, Pittsburgh, PA, 2014, pp. 2233-2240.

[4] **E. Gurpinar**, D. De, A. Castellazzi, D. Barater, G. Buticchi and G. Francheschini, "Performance analysis of SiC MOSFET based 3-level ANPC grid-connected inverter with novel modulation scheme," *2014 IEEE 15th Workshop on Control and Modeling for Power Electronics (COMPEL)*, Santander, 2014, pp. 1-7.

[3] Jianfeng Li, **E. Gurpinar**, S. Lopez-Arevalo, A. Castellazzi and L. Mills, "Built-in reliability design of a high-frequency SiC MOSFET power module," *2014 International Power Electronics Conference (IPEC-Hiroshima 2014 - ECCE ASIA)*, Hiroshima, 2014, pp. 3718-3725.

[2] D. Barater, **E. Gurpinar**, G. Buticchi, C. Concari, D. De, A. Castellazzi, "Performance analysis of UniTL-H6 inverter with SiC MOSFETs," *2014 International Power Electronics Conference (IPEC-Hiroshima 2014 - ECCE ASIA)*, Hiroshima, 2014, pp. 433-439.

[1] **E. Gurpinar**, S. Lopez-Arevalo, J. Li, D. De, A. Castellazzi and L. Mills, "Testing of a lightweight SiC power module for avionic applications," *Power Electronics, Machines and Drives (PEMD 2014), 7th IET International Conference on*, Manchester, 2014, pp. 1-6.

Patents

[2] **E. Gurpinar**, S. Chowdhury, "Power module with organic layers", US Patent 11,527,456 B2.

[1] S. Chowdhury, R. Sahu, **E. Gurpinar**, B. Ozpineci "High-performance capacitor packaging for next generation power electronics", US Patent 12,100,560 B2.

Patent Applications

[8] S. Chowdhury, **E. Gurpinar**, B. Ozpineci, "Power module", US 18/374,410, 2024

[7] V.P.G. Nagabhushana, J. Pries, **E. Gurpinar**, S. Chowdhury, "Method to suppress switching transition voltage oscillation in power electronics circuits by turn-off or turn-on current zero placement", US 18/588,785, 2024

[6] **E. Gurpinar**, "High Performance Redundant Liquid Cooling for Power Modules", US 18/793,039, 2024

[5] M. Mohammad, R. Wiles, **E. Gurpinar**, O. Onar, S. Chowdhury, J. Wilkins, "Only-stationary-side compensation network", US 18/305,713, 2023.

[4] M. Mohammad, R. Wiles, **E. Gurpinar**, O. Onar, S. Chowdhury, J. Wilkins, "Rotary transformer with integrated power electronics", US 18/305,734, 2023

[3] M. Mohammad, R. Wiles, **E. Gurpinar**, O. Onar, S. Chowdhury, J. Wilkins, "Rotor current prediction in an electric motor drive having an only-stationary-side compensation network", US 18/305,725, 2023

[2] S. Chowdhury, T. Raminosoa, B. Ozpineci, R. Wiles, **E. Gurpinar**, G.J. Su, J. Pries, "Electric drive system", US 17/733,251, 2023

[1] **E. Gurpinar**, D. De, A. Castellazzi, "Power Converter", WO2017089821A1, 2015.

Public Reports

[3] **E. Gurpinar**, et al. Failure Modes and Effects Analysis for Wireless and Extreme Fast Charging. No. DOT HS 813 137. United States. Department of Transportation. National Highway Traffic Safety Administration, 2021.

[2] **E. Gurpinar**, ”Highly Integrated Power Module”, US Department of Energy Vehicle Technologies Office Annual Progress Report, 2019,2020,2021

[1] **E. Gurpinar**, ”Advanced Multiphysics Integrations and Designs”, US Department of Energy Vehicle Technologies Office Annual Progress Report, 2018

Book Chapter

E. Gurpinar and B. Ozpineci, ”Emerging Packaging Concepts and Technologies”, SiC Power Module Design and Assembly: Building in Performance, Robustness and Reliability, IET, 2022.

References

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